SMART URBAN WATER DEMAND FORECASTING

A machine-learning study using a reproducible synthetic daily dataset

COURSE	BUILD AND DEPLOY FOR MACHINE LEARNING (BDML)
PROJECT	Daily municipal water-demand forecasting and model evaluation
REGISTRATION	241501185
DATA PERIOD	1 Jan 2022 - 31 Dec 2025 (synthetic; 1,461 daily records)

Project scope

This project demonstrates a complete BDML workflow: problem framing, synthetic data generation, chronological holdout validation, feature engineering, regression baselines, evaluation, interpretation, limitations, and reproducible code. The dataset is simulated for coursework; it is not a utility's operational data and must not be represented as observed city measurements.

Prepared for academic demonstration

1. Abstract

Urban water utilities need short-horizon demand estimates to plan pumping, storage, and daily operations. This project builds an interpretable regression model using weather, calendar, seasonal, and trend variables. A deterministic synthetic dataset contains 1,461 daily observations from 2022-2025. The final 20% of days is reserved for chronological testing. Ordinary least squares achieves a held-out MAE of 123.6 litres/day, RMSE of 155.6 litres/day, and R-squared of 0.860; this improves on a training-mean baseline MAE of 416.4 litres/day. Because the data are simulated, these scores validate the implementation and evaluation workflow, not real-world utility performance.

2. Problem definition and objectives

Given date, temperature, rainfall, humidity, weekend/holiday indicators, and a time index, estimate daily total water demand in litres. Objectives: (1) create a transparent, repeatable dataset; (2) establish a simple baseline; (3) train and evaluate regression models without random temporal leakage; and (4) document operational caveats.

This is a supervised regression problem. The target is continuous demand_litres. The intended prediction horizon is one daily record; lagged demand is deliberately excluded in the primary model because operational lag features require a defined forecast issue time and availability policy.

3. Dataset and feature design

The dataset generator uses a fixed random seed and produces 1,461 dates from 1 January 2022 through 31 December 2025. Each row includes temperature_c, rainfall_mm, humidity_pct, is_weekend, is_holiday, and demand_litres. The target is generated from a documented relationship with weather, calendar effects, a gradual trend, previous-day/previous-week demand and random noise. Lagged demand informs simulated target generation only; it is not passed to the fitted model.

Feature set: temperature, rainfall, humidity, weekend, holiday, normalized time index, and sine/cosine day-of-year terms. Seasonal encodings represent annual periodicity without an artificial discontinuity at year end. Numeric features are standardized using training-only statistics.

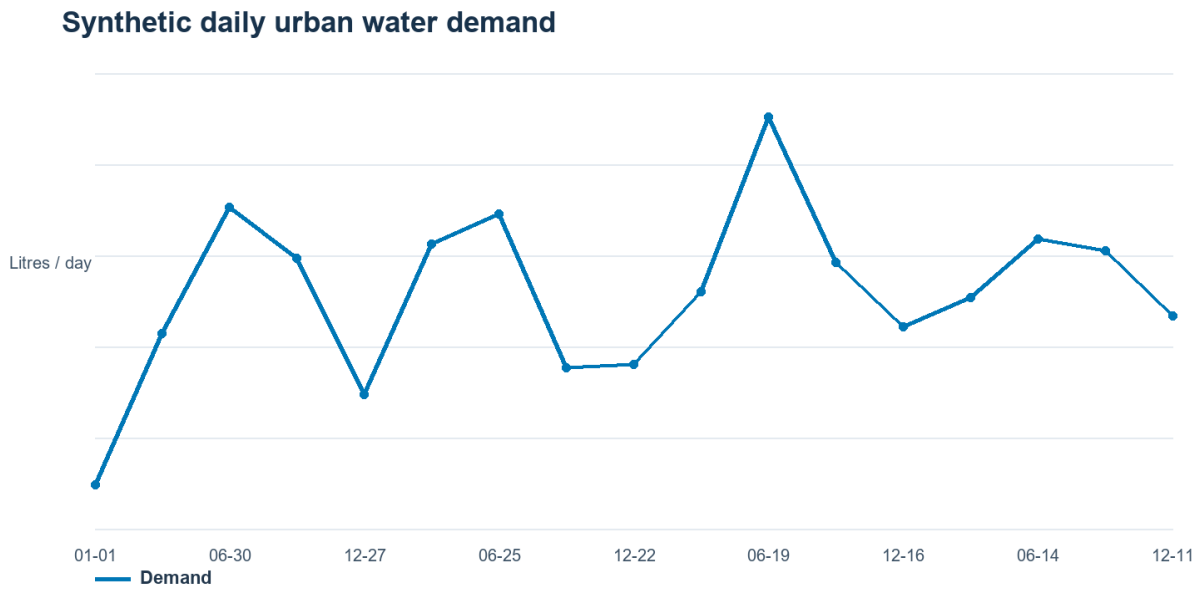


Figure 1. Sampled daily demand series across the generated four-year period. The values are synthetic.

4. Methodology

Data are sorted by date. The first 80% of observations (1,168 days) is used for fitting; the final 20% (293 days) forms the untouched chronological test set. The preprocessing mean and standard deviation are estimated only on training rows. Three comparisons are reported: a training-mean constant baseline, ordinary least squares, and ridge regression (L2 penalty = 4, intercept unpenalized). The primary model is ordinary least squares for interpretability.

Metrics: MAE is the average absolute error in litres/day; RMSE penalizes large misses more strongly; R-squared measures variance explained relative to the test-set mean. No hyperparameter tuning is performed, avoiding a misleading search on a small synthetic test set.

5. Results

Model	Test MAE (L/day)	Test RMSE (L/day)	Test R-squared
Training-mean baseline	416.4	-	-
Ordinary least squares	123.6	155.6	0.860
Ridge regression (alpha=4)	124.0	-	-

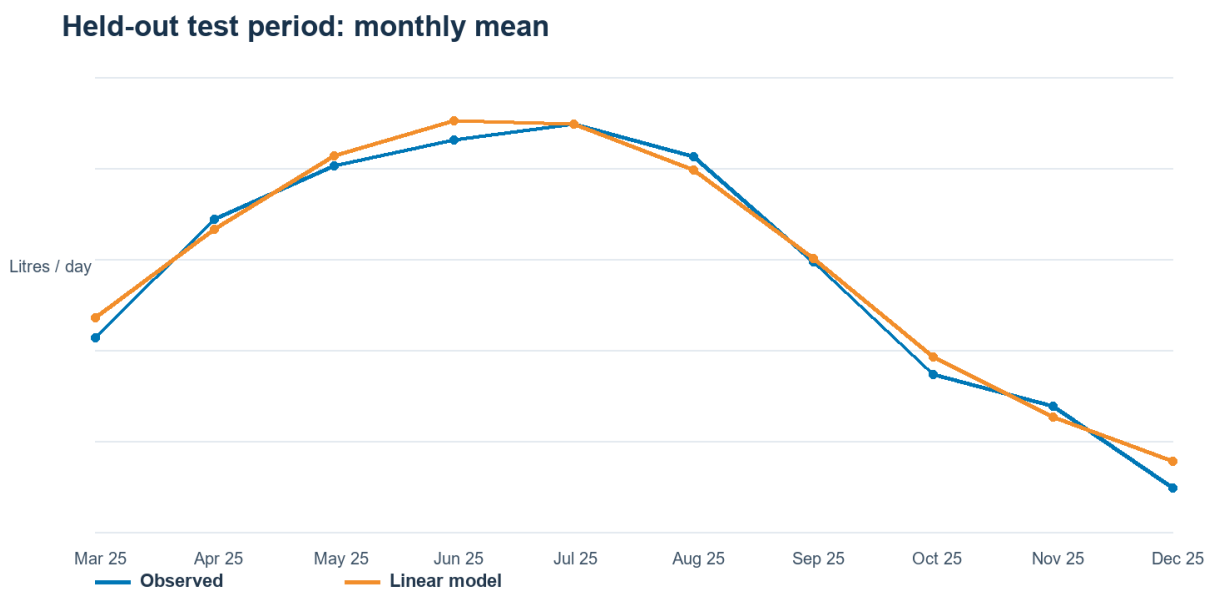


Figure 2. Observed and modelled monthly average demand on the held-out test period.

6. Error analysis and interpretation

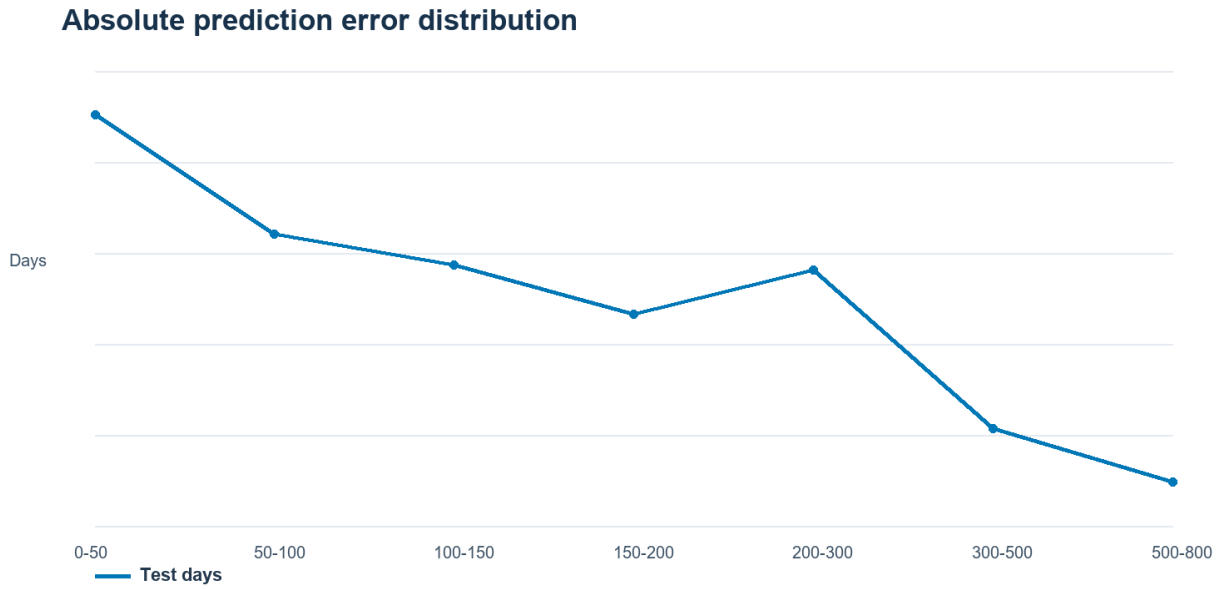


Figure 3. Absolute daily error bins on the chronological test set.

The model reduces MAE by 70.3% versus the constant baseline on this generated test segment. The test metrics should be read alongside the error distribution, not as a guarantee for an actual city. Strong performance is expected because the data-generation process deliberately encodes smooth relationships that are also represented in the feature set.

The normalized linear coefficients are included in the reproducible script's fitted parameter vector. Their signs and magnitudes reflect the synthetic generator and feature scaling. They are not causal effects and should not be interpreted as policy estimates.

7. Deployment concept

A real deployment would ingest weather forecasts and calendar data available at a fixed daily cutoff, validate units and missingness, generate a prediction with a timestamp and model version, and publish the estimate with an uncertainty band. Monitoring should track MAE by season, peak-day underprediction, data drift, and missing-input rates. Predictions should support—not replace—operator judgement and water-quality/safety procedures.

8. Limitations and responsible use

All records are synthetic and do not represent a named city or utility. No external source data or personal data are used. The generator is simplified: it omits population, tariffs, leakage, industrial loads, service interruptions, water restrictions, spatial differences, and extreme weather. Its target formula creates an intentionally learnable relationship, so test performance is not evidence of field readiness. A real study requires documented utility data permissions, privacy review, season-aware validation across multiple years, uncertainty estimates, and comparison with domain baselines.

9. Conclusion

The work demonstrates a reproducible, leakage-conscious regression pipeline for daily demand forecasting. Ordinary least squares is compared with a transparent baseline and ridge regression on a chronological holdout. The next research step is to repeat the same evaluation on authorized operational data and test whether weather and lagged demand add value over seasonal-naive forecasts.

10. Reproducibility

The repository includes the deterministic CSV, analysis code, report, and original project figures. The CSV is the analysis input; the script does not silently regenerate or modify it. Run `python -m pip install -r requirements.txt` followed by `python run_analysis.py` from the mini_project directory. Reported numbers are computed from the included CSV using a fixed chronological split; the script prints MAE, RMSE, R-squared, and baseline MAE.

Reproduction checklist

1. Keep the supplied CSV unchanged and confirm it contains 1,461 rows plus the header.

2. Install the single listed dependency (NumPy) in an isolated Python environment.

3. Run the analysis script; it parses dates, recreates the seasonal features and uses the first 80% of dates for training.

4. Compare the printed metrics with the results table. Small differences can occur across numerical library versions.

5. Do not use the synthetic-data scores to size infrastructure or claim field accuracy.

Repository file map

Path	Purpose
urban_water_demand.csv	Deterministic synthetic daily input table
run_analysis.py	Feature engineering, model fit, chronological test metrics
requirements.txt	Minimal runtime dependency list
assets/*.png	Original figures embedded in this report

All plotted figures are generated from the included synthetic observations or model predictions. The data use a fixed pseudorandom seed and contain no real customer records. Reproducing this educational exercise is not equivalent to validating a production forecasting system.

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